

LSM6DSV Family

Best-in-class, low noise IMU



LSM6DSV16BX for TWS and wearable



Sensors for the OnLife era



3D head tracking demo





LSM6DSV Family

Best-in-class low noise and high stability 6x IMU

Advanced processing in the edge
FSM and MLC 2.0; ASC(1) with ST libraries!

Low noise accelerometer and gyro
60 $\mu\text{g}/\sqrt{\text{Hz}}$ (All FS), 2.8 $\text{mpds}/\sqrt{\text{Hz}}$

Embedded sensor fusion
Low power, low latency – high performance





LSM6DSV & LSM6DSV16X

6x IMU for UI, OIS & EIS

High end IMUs with OIS/EIS channel, exportable AI & Qvar™



Key features

- Dedicated core UI, OIS & EIS
- HAODR < 1% (LSM6DSV16X)
- Finite State Machine
- MLC 2.0 and Qvar (LSM6DSV16X)
- VDDIO: 1.2V
- I2C / SPI / I3C (1.1) primary for UI and QVAR

LSM6DSV16X
(2.5x3.0x0.86mm3 LGA14)

High End IMU low Noise		
Accelerometer	FS	±16g
	UI noise	60 µg / √Hz
Gyroscope	FS	±4000 dps
	UI Noise	2.8 mdps / √Hz
	BI	3dph
Low Power	UI	0.65 mA

Target applications

- Motion tracking and gesture detection
- AR / VR / MR applications
- Wearables & IoT connected devices
- Indoor navigation
- Smartphones and handheld devices
- EIS and OIS for camera applications





LSM6DSV32X

IMU for sport, wearable & IoT application with 32g accelerometer & MLC + Qvar™



LSM6DSV32X
(2.5x3.0x0.86mm3 LGA14)

- ✓ FS: ± 4000 dps, ± 32 g
- ✓ Idd 0.65 mA
- ✓ Gyro noise 2.8mdps/ $\sqrt{\text{Hz}}$
- ✓ G_OffDr: ± 0.008 dps/ $^{\circ}\text{C}$

IMU with Accelerometer up to 32g

Configurable Power mode

Smart feature configurable

Advanced interfaces: I2C, SPI & I3C® 1.1

Activity recognition low power (MLC) & Qvar™



LSM6DSV16BX for TWS and wearable

Unique IMU for user interface & bone conduction detection



Low noise, wide & flat bandwidth accelerometer
 $30 \mu\text{g}/\sqrt{\text{Hz}}$; THD+N <1% ; BW > 1 kHz

Dedicate interface for bone conduction detection
TDM interface for data synch

Highly integrated, low power IMU
U/I + TDM : 0.95 mA (-70% vs discrete solution)

LSM6DSV16BX for TWS and AR / VR

Activities & gesture recognition



Always-on process in the edge

- Activities recognition & context awareness
- Carry position

Audio 3D & Spatial Audio



Immersive audio experience

- Accurate 3D Audio with continuous monitoring by SFLP
- Wide BW, Low Noise for Speech enhanced

User Interface



Motion tracking & user interface

- Continuous tracking low power
- Voice Activity Detection
- Immersive experience with *touch, long press & swipe* by Qvar

Sensors for the OnLife era

Standalone devices able to sense, process and take action



Embedded sensor fusion with game rotation vector

MLC: In-sensor classification engine based

Adaptive self configurability

Sensor fusion low power for always-on applications

SFLP plug & play for processing in the edge



Features

- 6x Game Rotation Vector (accelerometer + gyroscope)
- Rotation vector output available as quaternion format (W, X, Y, Z)
- X, Y, Z quaternion components can be stored in FIFO

Performance

- Static accuracy (heading, pitch, roll) 0.5⁽¹⁾, 1.5, 1.5 deg
- Low Dynamic accuracy (heading, pitch, roll) 0.7⁽¹⁾, 0.5, 0.5 deg
- Calibration time⁽²⁾ 0.8 s
- Orientation stabilization time 0.7 s

Power consumption

- SFLP extra power: 30uA @120Hz in the edge with 50% power reduction vs. external MCU⁽³⁾ processing

(1) Heading / 5min

(2) Time required to reach steady state

(3) Same Sensor Fusion software library running on STM32L476RG cortex M4 @ 65 uA (120Hz ODR)

Gaming



Wearable

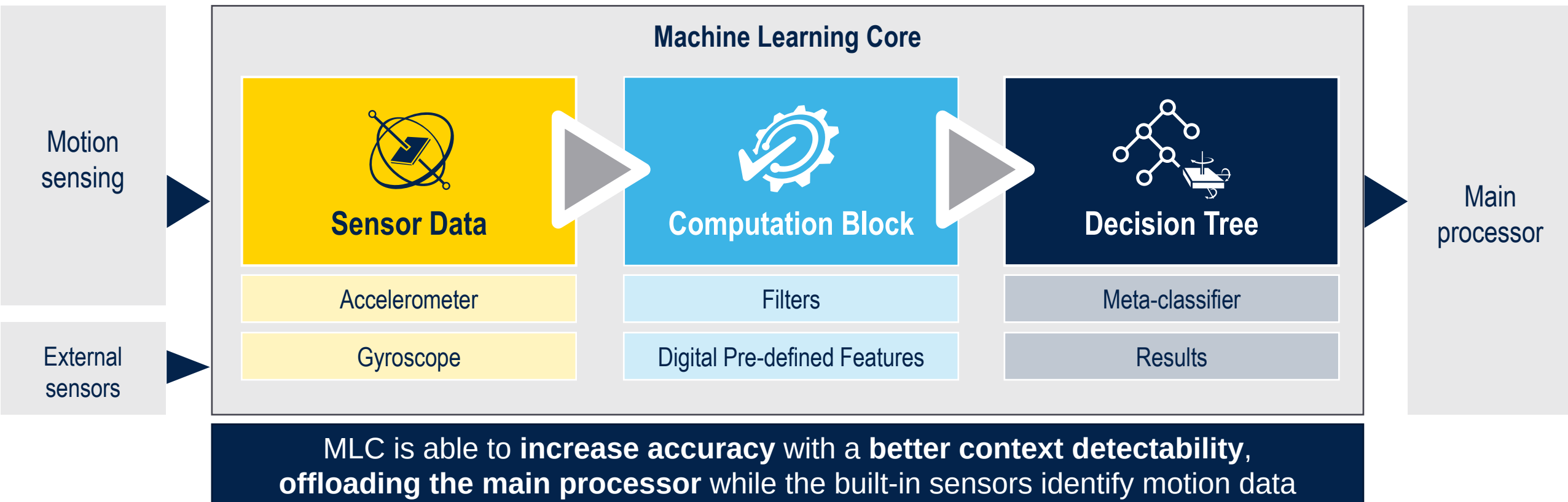


Navigation



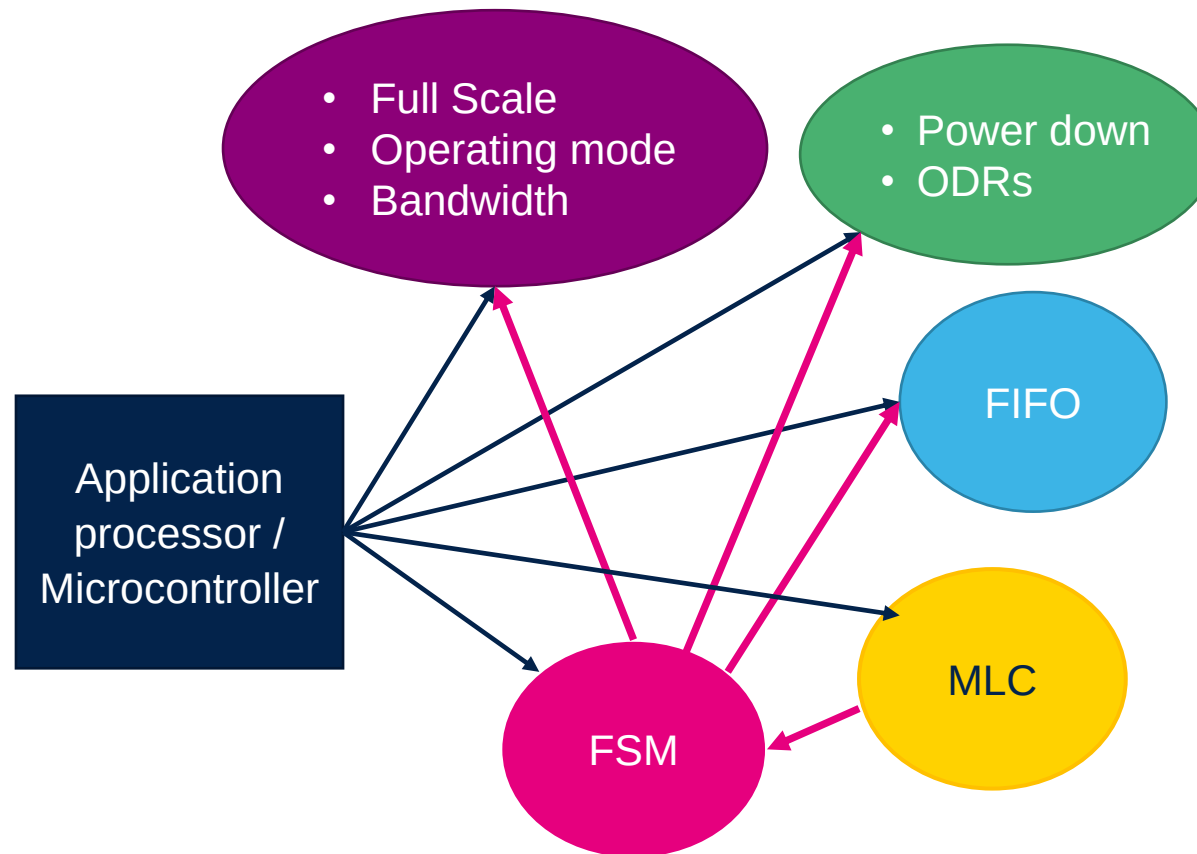
Machine Learning Core

MLC is an in-sensor classification engine based on a decision tree logic



Adaptive self-configuration

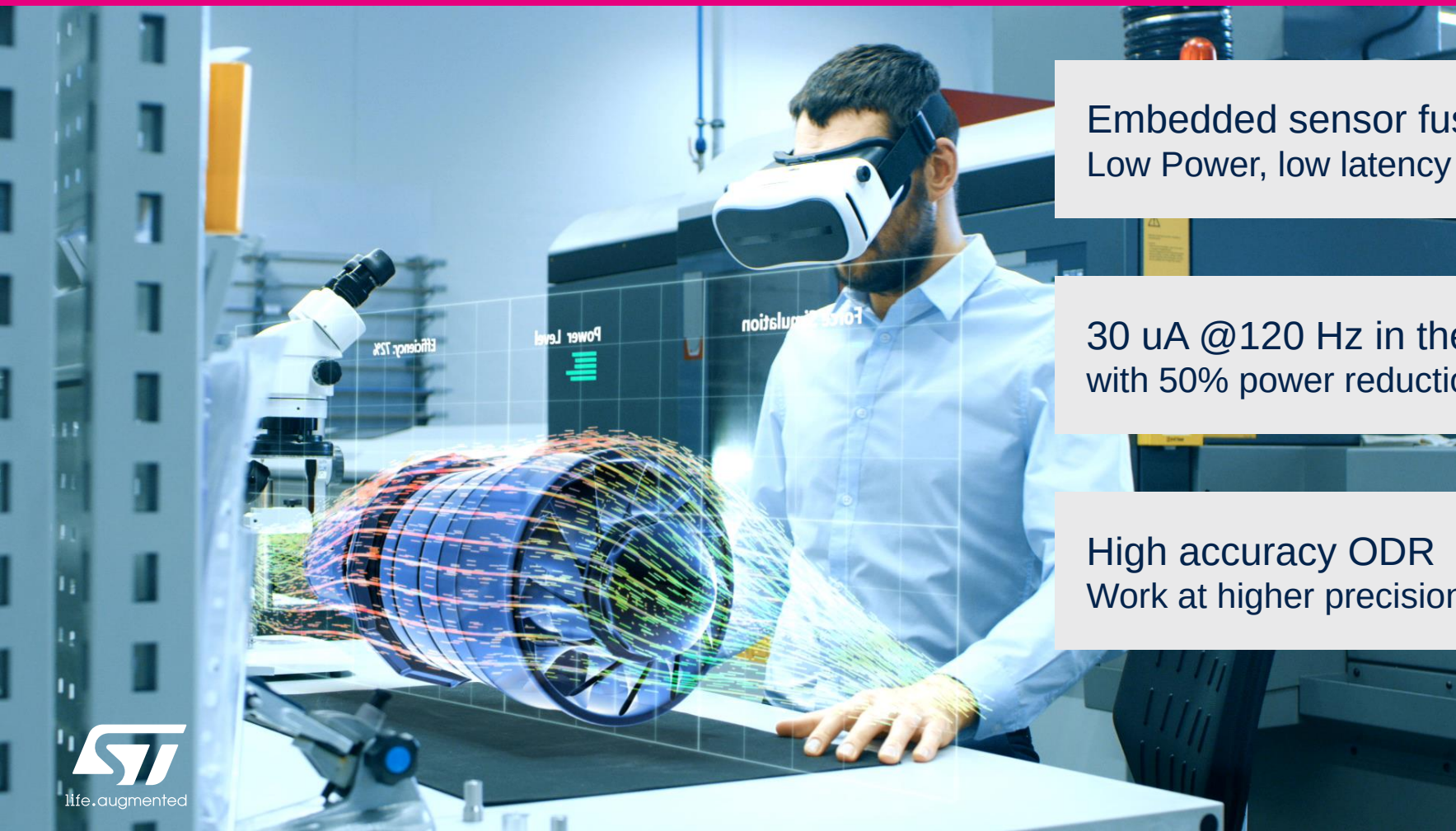
**FSM can change the device configuration:
Application processor can be kept in sleep mode (power optimization)**



ODR = Output Data Rate
FIFO = First In First Out
MLC = Machine Learning Core
FSM = Finite State Machine

3D head tracking demo

Enabling AR / VR headsets and TWS devices



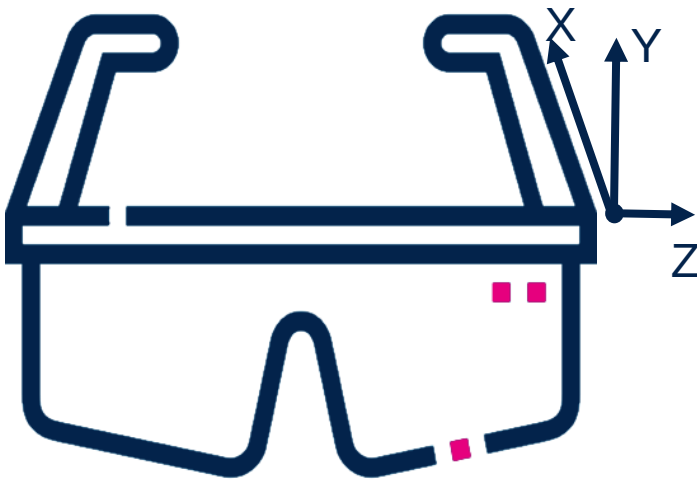
Embedded sensor fusion
Low Power, low latency – high performance

30 μA @120 Hz in the edge
with 50% power reduction vs. external MCU processing

High accuracy ODR
Work at higher precision, when it is needed

3D head tracking

Based on SensorTile.box STEVAL-MKSBOX1V1
custom version: LSM6DSOX → LSM6DSV16X



Motion Sensors



High-Performance 6-axis IMU, with
Machine Learning Core

LSM6DSV16X



High-performance and low power
accelerometers

LIS3DHH & LIS2DW12



Magnetometer

LIS2MDL

Environmental Sensors



Altimeter / Pressure sensor

LPS22HH



Accurate temperature sensor

STTS751



Analog wide-band microphone

MP23ABS1

